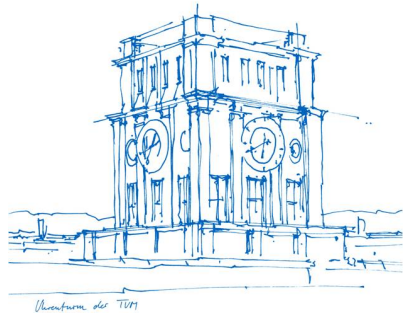


PoreAccess-CO₂: From Random Packing to an Interpretable Transport Closure

Structure-resolved diffusion and adsorption in
DEM-generated zeolite beds

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Presentation outline

Motivation and research question

Model scope and physical mechanisms

Numerical methods and diffusivity prediction

Results and robustness

Conclusions and next steps

Section 1 of 5

Motivation and research question

Why can similar beds adsorb differently?

CO₂ adsorption matters and packing structure is a missing scale

Why CO₂ capturing matters?

- Food systems contribute about **one-third** of global anthropogenic greenhouse-gas emissions.
- Food and bioprocessing include recoverable biogenic CO₂ streams, for example from fermentation.
- Regenerable sorbents offer a compact capture and purification route.

What is already established?

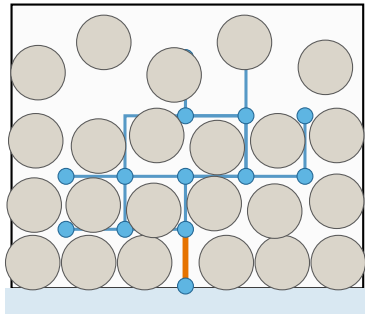
- CO₂ equilibrium and uptake kinetics on zeolite 13X are widely studied.
- Fixed-bed models normally use bulk porosity and a fitted effective transport coefficient.
- Random inlet-connected paths remain hidden inside that coefficient.

Novelty: connect random particle geometry to a fast bed-scale uptake model through a physically calculated pore-accessibility descriptor.

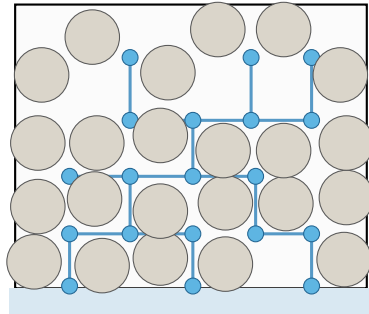
Same porosity can hide different inlet-connected pathways

Can a pore-accessibility measure predict bed-scale transient transport better than porosity alone?

Packing A: inlet bottleneck



Packing B: multiple inlet paths



- pore body
- diffusive throat
- zeolite bead
- inlet bottleneck

Same beads and similar porosity; different numbers and conductances of inlet-connected paths.

Why zeolite 13X and what does the study isolate?

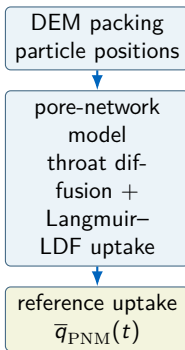
Why zeolite 13X?

- established benchmark sorbent for CO₂
- strong equilibrium uptake at low partial pressure
- commercial millimetre-scale beads form random beds

Controlled model problem

- dry, isothermal, stagnant gas
- CO₂ enters from a reservoir below the bed
- pore-network model resolves throat diffusion and bead adsorption
- no imposed flow; no CFD

How strongly does random void connectivity change access to the adsorbent?



Section 2 of 5

Model scope and physical mechanisms

From particle arrangement to diffusion and adsorption

Twenty beds isolate variability caused by random arrangement

Quantity	Fixed value
beads per bed	1850
bead diameter	1.6 mm
tube diameter	16 mm
D_{tube}/d_p	10
temperature	298.15 K
inlet c_{CO_2}	6.05 mol m ⁻³ *
simulation time	5000 s

Only the DEM insertion seed changed

- Delaunay–Voronoi networks were built directly from particle coordinates
- all networks were connected and spanning
- quasi-uniform Sobol sampling allocated pore volumes without a voxel grid
- transient and 1D models passed mass-balance checks

* Approximately 15 vol.% CO₂ at atmospheric pressure and 298 K.

The ensemble tests arrangement effects under otherwise identical conditions.

Pore-network geometry and throat diffusion

Network construction

- four bead centres define a Delaunay tetrahedron; its circumcentre is a pore body
- tetrahedra sharing a triangular face are joined by a throat
- topology comes from particle coordinates, not voxels
- 2^{22} Sobol points sample the solid and gas space.
- Gas-point counts allocate the calculated void volume to pores.

Diffusion through throat $i-j$

$$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}, \quad \dot{n}_{ij} = G_{ij}(c_j - c_i).$$

$L_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$ is the pore-centre distance. The irregular opening between three beads is approximated by an equivalent circular area

$$A_{ij} = \pi r_{ij}^2,$$

where r_{ij} is the clearance of the shared Delaunay face.

The circular throat area is a geometric modelling assumption; G_{ij} has units of $\text{m}^3 \text{s}^{-1}$.

Langmuir equilibrium and bead uptake

Local equilibrium loading

$$p_k = c_k RT, \quad q_k^* = \frac{q_s b p_k}{1 + b p_k}.$$

Finite-rate uptake

$$\frac{dq_k}{dt} = k_{\text{LDF}} (q_k^* - q_k).$$

- q_s : maximum CO₂ loading [mol/kg adsorbent]
- b : Langmuir affinity [Pa⁻¹], with $b p_k$ dimensionless
- p_k : local CO₂ partial pressure [Pa]
- q_k^* : equilibrium loading at p_k

Meaning of the rate equation

- q_k is the current loading of bead k
- k_{LDF} controls how quickly q_k approaches q_k^*
- gas removed from adjacent pores is added to the bead

Pore-network outputs

- pore concentrations $c_i(t)$
- particle loadings $q_k(t)$
- mean uptake $\bar{q}_{\text{PNM}}(t)$

Coupled diffusion and adsorption conserve the CO₂ inventory.

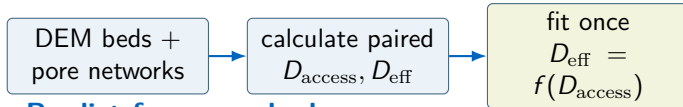
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Numerical methods and diffusivity prediction

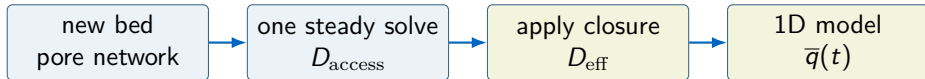
How we calculate accessibility and calibrate the reduced model

Development and later use are two different workflows

Develop closure



Predict for a new bed



For a new bed, the transient pore-network adsorption model is no longer required.

The homogeneous 1D uptake model

Gas balance in each axial cell

$$\varepsilon_b \frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t}.$$

Adsorbent loading in each axial cell

$$\frac{\partial q}{\partial t} = k_{\text{LDF}} \left[\frac{q_s b R T c}{1 + b R T c} - q \right].$$

- Divide the bed height into 100 equal slices, each spanning the tube cross-section.
- Solve gas concentration and mean solid loading together in each slice.
- Trial D changes c , which changes adsorption.

Average the calculated loading

$$\bar{q}_{1D}(t; D) = \frac{1}{H} \int_0^H q(z, t; D) dz.$$

For 100 equal axial cells:

$$\bar{q}_{1D}(t; D) = \frac{1}{100} \sum_{\ell=1}^{100} q_{\ell}(t; D).$$

- Match bed height, porosity and inlet condition to the network.
- Keep isotherm, kinetics and initial state fixed. Only D changes.

The 1D grid is separate from the pore network.

D_{access} and D_{eff} answer different questions

D_{access} : geometric accessibility

- fix concentration at inlet and interior plane
- solve one steady, linear diffusion problem
- no adsorption and no time integration

$$D_{\text{access}} = \frac{G_{\text{access}} H}{A_{\text{tube}}}.$$

G_{access} is total steady flux per concentration difference. H is bed height; A_{tube} is tube area.

D_{eff} : transient reduced-model coefficient

- full transient network gives $\bar{q}_{\text{PNM}}(t)$
- repeated 1D solves give $\bar{q}_{1\text{D}}(t; D)$

Define the curve residual

$$e_m(D) = \bar{q}_{1\text{D}}(t_m; D) - \bar{q}_{\text{PNM}}(t_m).$$

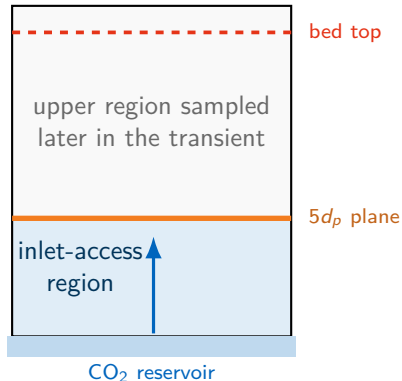
$$\text{NRMSE}(D) = \frac{\sqrt{\sum_m w_m e_m(D)^2}}{\bar{q}_{\text{PNM}}(t_f)},$$

$$D_{\text{eff}} = \arg \min_D \text{NRMSE}(D).$$

The trapezoidal weights w_m account for unequal time spacing and satisfy $\sum_m w_m = 1$.

D_{access} is cheap because it avoids adsorption dynamics and optimization.

Why use an interior plane?



A plane at $5d_p$ includes

- the inlet openings
- second-layer and deeper bottlenecks
- parallel paths connecting inlet to the plane

The bed top answers a different question

- measure conductance through the entire bed
- include upper bottlenecks not sampled early

Why not predict uptake directly from G_{access} ?

Direct endpoint relation

$$G_{\text{access}} \longrightarrow \bar{q}(5000 \text{ s})$$

- strong correlation: $\rho_s = 0.968$
- ranks beds or predicts one endpoint
- valid at the selected time and conditions

Transport closure

$$G_{\text{access}} \longrightarrow D_{\text{access}} \longrightarrow D_{\text{eff}} \longrightarrow \bar{q}(t)$$

- generates the complete uptake curve
- retains conservative 1D transport
- gives axial $c(z, t)$ and $q(z, t)$ profiles

If only uptake at 5000 s is needed, G_{access} is sufficient; D_{eff} is needed for a reusable transient model.

Diffusivity optimization and closure calibration

Fit one diffusivity per bed

1. Choose a trial D within prescribed bounds.
2. Solve the coupled 1D equations for $c(z, t)$ and $q(z, t)$.
3. Compare mean uptake with the network using time-weighted NRMSE.
4. Use previous errors to select the next D and repeat (optimization).

Learn the relationship across beds

- Pair each fitted D_{eff} with its geometry-based D_{access} .
- Fit the closure $D_{\text{eff}} = f(D_{\text{access}})$ on the training beds.
- Leave one bed out to assess diffusivity prediction.

The bounded search operates in $\log_{10} D$ and starts deterministically.

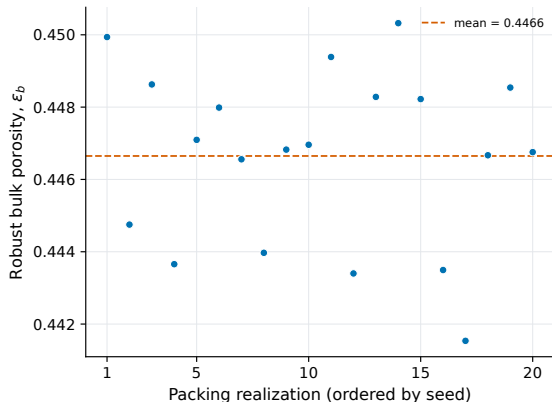
Per-bed optimization supplies the targets for closure calibration.

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Results and robustness

Uptake variability, predictive closure and depth sensitivity

The twenty beds have nearly equal bulk porosity



Bulk-porosity CV: 0.55%.

Controlled ensemble

- same bead count and diameter
- same tube geometry
- only the insertion seed changes

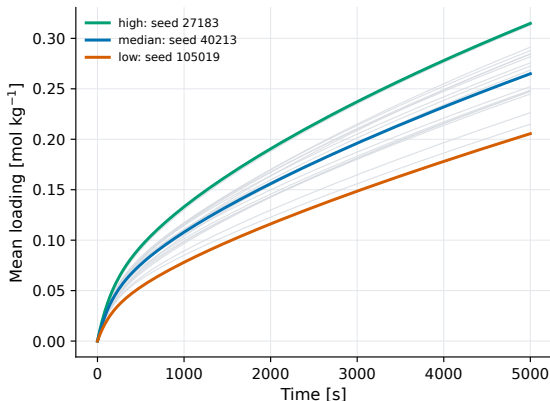
CV: coefficient of variation

$$CV_{\epsilon} = s_{\epsilon} / \bar{\epsilon}$$

s_{ϵ} : standard deviation

$\bar{\epsilon}$: mean porosity

Similar porosity still produces different uptake curves

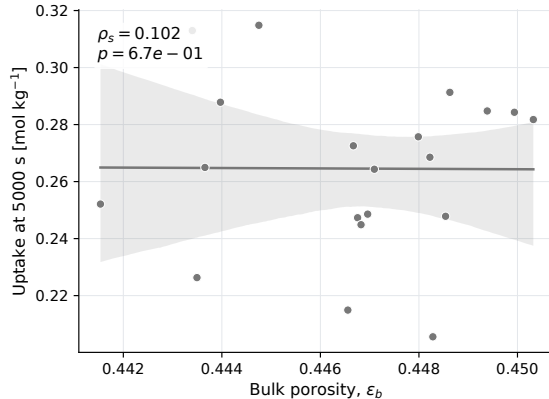


Network reference

- all twenty uptake curves
- highlighted beds span low, median and high final uptake
- identical adsorption parameters

At 5000 s: 0.206–0.315 mol kg⁻¹, with an uptake CV of 11.0%.

Bulk porosity alone does not explain final uptake

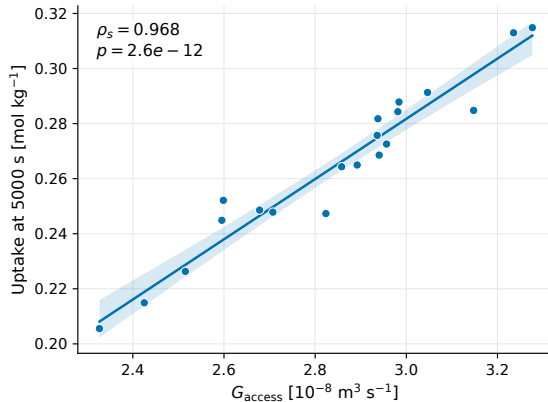


Porosity baseline

- each point represents one bed
- weak rank association
- wide uncertainty around the fitted relationship

Void volume alone does not describe access to the adsorbent.

Accessibility explains differences in final uptake



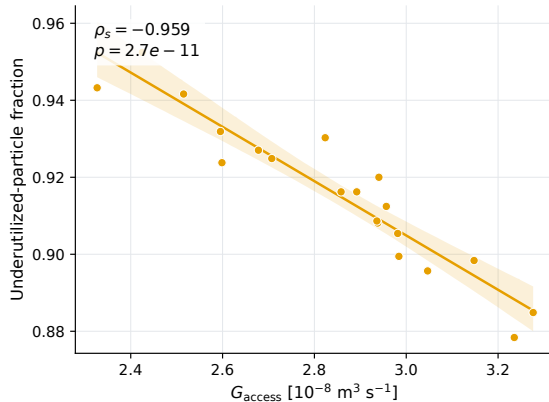
Steady network descriptor

- G_{access} measures inlet-to-interior conductance
- greater conductance accompanies greater uptake

Spearman $\rho_s = 0.968$.

Inlet-connected pathways explain variability that porosity misses.

Greater accessibility reduces particle underutilization



Higher G_{access} accompanies fewer underutilized particles.

Underutilized fraction

$$f_{\text{under}} = \frac{1}{N_p} \sum_{k=1}^{N_p} I_k.$$

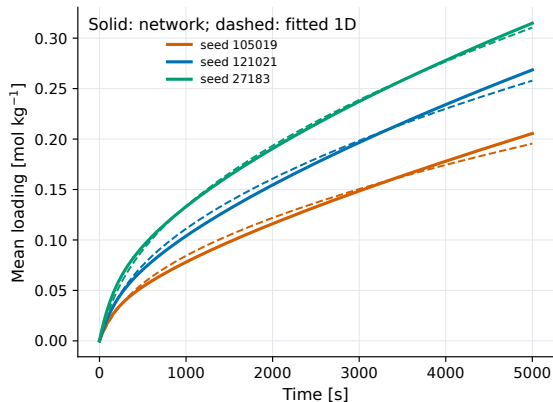
$I_k = 1$ when

$$q_k(t_f) < 0.5q^*(p_{\text{inlet}}).$$

$t_f = 5000$ s.

The indicator equals one for a particle below half the inlet-equilibrium loading.

One fitted D_{eff} reproduces each uptake curve



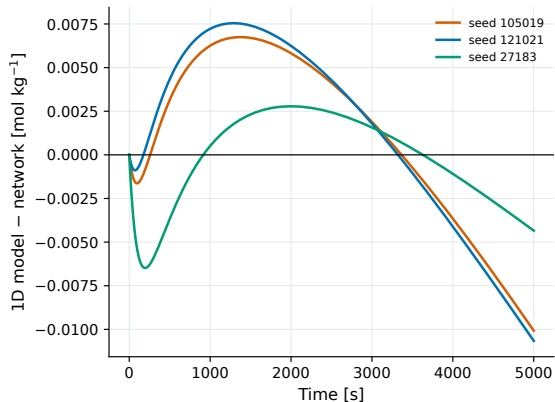
Per-bed optimization

1. solve the 1D model for a trial D
2. evaluate time-weighted NRMSE against the network curve
3. use previous errors to select the next D and repeat (optimization)

Solid: network.
Dashed: fitted 1D model.

Across twenty beds, the mean fitted-curve NRMSE is 1.58%.

Residuals show the remaining reduction error



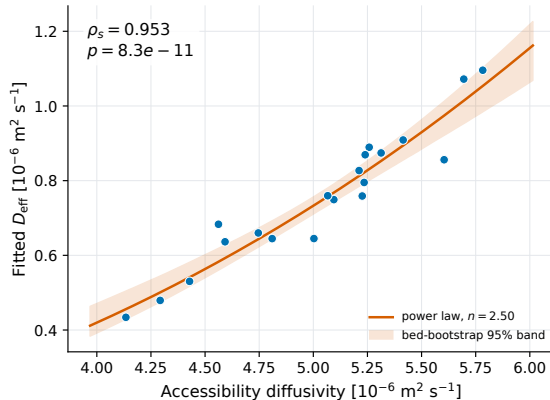
Curve residual

$$e(t) = \bar{q}_{1D}(t) - \bar{q}_{PNM}(t).$$

- positive: excess predicted uptake
- negative: insufficient predicted uptake

Time-weighted NRMSE across all beds: 0.84–2.46%.

A positive power law relates accessibility to diffusivity



Selected closure

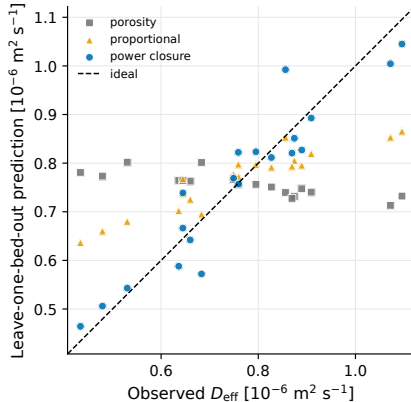
$$\frac{D_{\text{eff}}}{D_m} = C \left(\frac{D_{\text{access}}}{D_m} \right)^n.$$

$$D_{\text{access}} = \frac{G_{\text{access}} H}{A_{\text{tube}}}.$$

The shaded band shows bootstrap uncertainty in the fitted relationship.

The fitted exponent is $n \approx 2.50$ for this ensemble.

The closure predicts diffusivity for a withheld bed



Closures compared

$$D_{\text{eff}} = a_0 + a_1 \varepsilon_b,$$

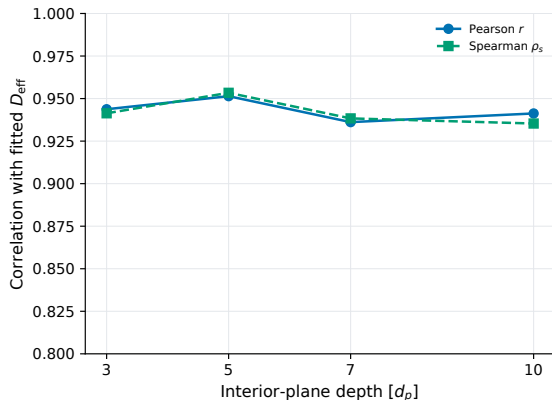
$$D_{\text{eff}} = \alpha D_{\text{access}},$$

$$\frac{D_{\text{eff}}}{D_m} = C \left(\frac{D_{\text{access}}}{D_m} \right)^n.$$

Fit on nineteen beds, then predict the remaining bed.

LOOCV R^2 : power law 0.890, proportional 0.555, porosity -0.234 .

Accessibility remains correlated across diagnostic depths



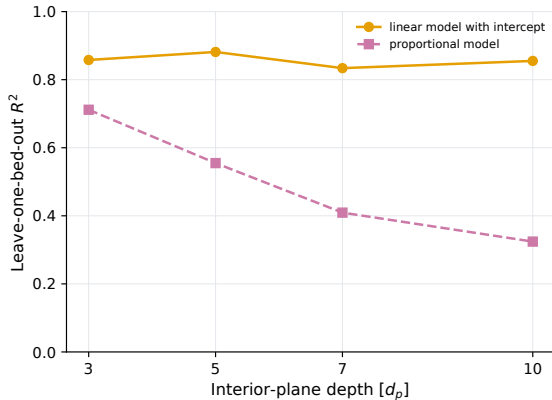
Depth sensitivity

- planes at 3, 5, 7 and $10d_p$
- Pearson: linear association
- Spearman: rank association

Depth is measured above each network's lowest pore.

Both correlations remain high across the tested depths.

Predictive performance is stable across diagnostic depths



Models in this comparison

- linear model with intercept

$$D_{\text{eff}} = aD_{\text{access}} + b$$

- proportional model

$$D_{\text{eff}} = \alpha D_{\text{access}}$$

The $5d_p$ plane gives the highest linear-model score in this ensemble.

The linear model with intercept remains predictive from $3d_p$ to $10d_p$.

Section 5 of 5

Conclusions and next steps

What the evidence supports and what remains to validate

What the present evidence establishes

Supported

- porosity alone misses important packing variability
- inlet-connected topology controls transient utilization
- one D_{eff} reproduces each detailed curve accurately
- D_{access} provides an interpretable predictive closure

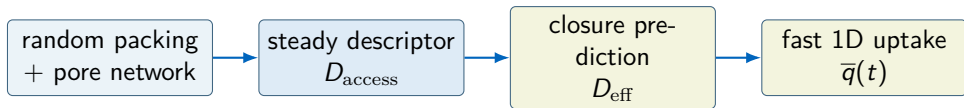
Not yet claimed

- industrial breakthrough performance
- effects of flow, heat or water competition
- transfer to other bead sizes or D_{tube}/d_p
- a universal power-law exponent

The contribution is a structure-to-model closure for a controlled diffusion-driven adsorption problem.

Pore accessibility retains what bulk porosity loses

Equal void volume does not imply equal transport. How that void space connects to the inlet matters.



20 beds
controlled ensemble

1.58%
mean reduction error

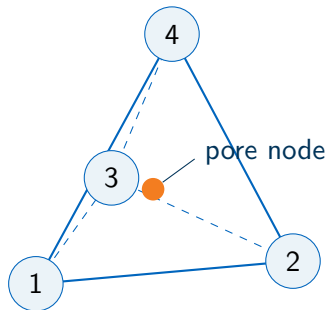
$R^2 = 0.890$
withheld-bed closure

Backup slides

How the pore network is constructed

Particle geometry, Sobol volume allocation and transport

1. Particle positions define candidate pore nodes



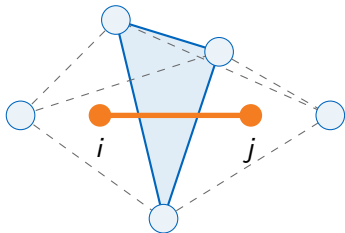
Schematic 3D projection
Bead centres form a tetrahedron

- Start with known bead centres, radii and bed boundaries.
- Delaunay tetrahedralization groups neighbouring centres into sets of four.
- Locate the circumcentre: the point equidistant from those four centres.
- Retain valid points inside the bed and outside the solid as pore nodes.

A node represents a gas region. Its volume is assigned later.

The particle geometry determines where the pore nodes are placed.

2. Shared particle faces define pore connections



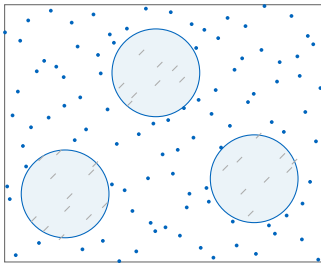
Two tetrahedra share three beads
Schematic 3D projection

- Two neighbouring tetrahedra share a triangular face.
- The three beads around that face define an opening.
- A valid opening connects pore nodes i and j through a throat.
- Estimate throat length L_{ij} and circular area A_{ij} from geometry.

$$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}.$$

At this stage we know the nodes and connections. Sobol sampling has not been used.

3. Sobol points measure the gas space



Illustrative 2D Sobol sample

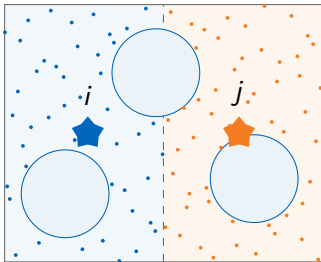
Blue: gas points; grey crosses: solid points

- Distribute $2^{22} \approx 4.2$ million sample points through the cylindrical bed.
- Sobol sampling covers the domain with few gaps and clusters.
- Check each point against the bead surfaces.
- Discard points inside solid particles and retain gas-space points.

These are measuring points, not extra pores or mesh cells.

Sobol sampling estimates volume. It does not generate network connections.

4. Assign gas-space samples to the nearest pore



Two illustrative pore nodes
Each gas point takes its nearest node's colour

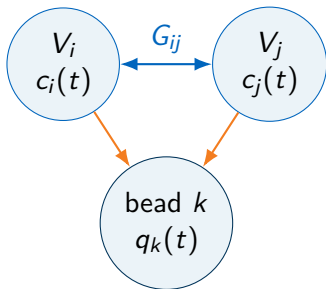
- Assign every retained gas point to its nearest pore node.
- Count the assigned points: N_i for pore i .
- A larger count gives a larger share of the calculated void volume.

$$V_i = V_{\text{void}} \frac{N_i + 0.5}{\sum_j (N_j + 0.5)}.$$

- The small 0.5 correction prevents zero volumes.
- Normalization ensures $\sum_i V_i = V_{\text{void}}$.

The total volume is conserved, while each individual pore volume is an estimate.

5. The network carries coupled diffusion and adsorption



Schematic during net adsorption

- Pore volumes store gas: $n_i = V_i c_i$.
- Throats transfer gas according to concentration differences.
- Associated beads adsorb gas from adjacent pores.
- Boundary connections supply gas from the inlet reservoir.

$$\dot{n}_{j \rightarrow i} = G_{ij}(c_j - c_i), \quad \frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k).$$

Throats control gas movement. Pore volumes control local gas storage.